

LAB NO: 10

LAB TITLE : More About Functions in C++

OBJECTIVE :

The objective of this lab is to understand the concept of functions in C++. This lab explains function call by value, function call by reference, and default parameters. It also helps students learn how functions make programs easier, shorter, and more organized.

THEORY :

A function is a separate block of code used to perform a specific task in a program. Functions reduce repetition in programming and improve readability of code. In C++, functions can receive values from the main program and can also return values after processing.

Example :

```
#include <iostream>
using namespace std;

void displayMessage()
{
    cout << "Welcome to Functions";
}

int main()
{
    displayMessage();

    return 0;
}
```

Function Call by Value :

In call by value, a copy of the variable is sent to the function. Changes made inside the function do not affect the original variable in the main function.

Syntax ;

```
functionName(variable);
```

Function Call by Reference :

In call by reference, the original variable is passed to the function using the **&** symbol. Any changes made inside the function also change the original value.

Syntax :

```
void functionName(int &variable);
```

Default Parameters :

Default parameters are values already assigned in the function declaration. If the user does not provide values during function calling, the default values are used automatically.

Syntax:

```
int add(int a = 5, int b = 2);
```

SOFTWARE :

1. DEV-C++ IDE.
2. C++ Compile .

LAB TASKS :

Function with Parameters

1. Declare a function called "multiply" that takes two integer parameters:"num1"and"num2".
2. Inside the function, calculate the product of the two numbers.
3. Display the result within the function.
4. Call thefunction from the main program, passing two numbers, to perform the multiplication.

```
#include <iostream>
using namespace std;

void calculateProduct(int x, int y) // Ye function do numbers ko multiply
kare ga
{
    int result;
```

```

        result = x * y; // Dono numbers ka product calculate karna

        cout << "Product = " << result << endl; // Result display karna
    }

int main()
{
    int num1 = 5;
    int num2 = 4;

    calculateProduct(num1, num2); // Result display karna

    return 0;
}

```

Function with Return Value

1. Declare a function called "getSquare" that takes an integer parameter: "number".
2. Inside the function, calculate the square of the number.
3. Return the result.
4. Call the function from the main program, passing a number, and display the square.

```

#include <iostream>
using namespace std;

int calculateSquare(int value) // Ye function number ka square return kare ga
{
    return value * value; // Square return karna
}

int main()
{
    int number = 6;

    int result = calculateSquare(number); // Function call karna
}

```

```
    cout << "Square = " << result << endl; // Result display karna

    return 0;
}
```

CONCLUSION :

In this lab, we learned different types of functions in C++. We understood function call by value, function call by reference, and default parameters. We also learned how to create functions with parameters and return values.